flashjet

GPU-native jet clustering with the full merge history

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Benchmarks: **ATLAS Open Data** (CC0). Closures: analytic predictions on toy showers.

Summary — what this talk shows

A GPU implementation of the generalised-k_t algorithms (anti-k_t, k_t, Cambridge–Aachen) that clusters batches directly in device memory and keeps the **complete merge history** — so substructure follows as cheap array reads, not a second clustering pass.

Validation ladder, each rung closed against an independent reference:

rung	reference	headline
unit tests	independent NumPy tree-walks	129 passed, incl. GPU paths
analytic closures	leading-log predictions, toy shower	z_g on the $1/z$ curve; areas; β-ordering
vs FastJet	same events, same job	**150/150** points, **100 %** n_{jets} agreement
vs the experiment	ATLAS's own reconstructed jets	**100 %** match, ∼600 clusters/event
speed	vectorised FastJet, identical input	**39–99**×, growing with multiplicity

flashjet reproduces FastJet — and the experiment's own jets — while running roughly two orders of magnitude faster than per-jet CPU clustering. It is a **prototype**: validated, but not yet inside any experiment's software.

What flashjet is

Clustering is the CPU step in a GPU pipeline

Jet clustering sits on the critical path of nearly every jet-level ML pipeline — and is almost always done **off the accelerator**.

Four-momenta are copied to the host, clustered, copied back.

That round trip dominates when

- training revisits the same events for **many epochs**,
- jets are **reclustered** under many variations,
- clustering must sit **inside** a differentiable pipeline.

flashjet

Implements anti-k_t, k_t and C/A **natively on the GPU**.

Data never leaves device memory during clustering.

Triton/PyTorch. One entry point; runs unchanged on CPU and GPU.

It returns the tree, not just the jets

Clusters a **batch of jets or events at once**, and returns **two** things:

1. the particle → jet assignment;
2. the **complete merge history** — which pair merged, into what, at what distance.

Why the history matters

Groomed mass, z_g, R_g, Lund coordinates, exclusive subjets become **reads of that tree**.

No second clustering pass.

`jet_idx`, `n_jets`	particle-to-jet assignment
`hist_p1`, `hist_p2`, `hist_child`	which pair merged into what
`hist_d`	the d_{ij} at each merge step

One entry point

```
out = flashjet.cluster(p4, mask, R=1.0, algorithm="antikt")
#   p4        (B, N, 4) torch tensor   |  mask  (B, N) bool
#   algorithm "antikt" | "kt" | "cambridge"

out.n_jets                                          # jets found
out.jets_p4(p4)                                     # jet four-momenta
out.exclusive_jets(n_jets=3)                        # F1
out.groomed_jets(p4, R, z_cut=0.1, beta=0.0)        # F2  soft drop / mMDT
out.lund_coordinates(p4, R)                         # F3  primary Lund plane
```

Same call on CPU and GPU. `jets_p4` is a `scatter_add` of constituents, so **gradients flow through the clustering**.

What is implemented

Clustering

- generalised-k_t: anti-k_t, k_t, C/A
- batched, ragged input via a mask
- three backends, auto-dispatched by size
- merge history recorded in-kernel
- jet regime *and* event regime

Substructure — reads of the history

- **F1** exclusive k_t subjets
- **F2** soft drop / mMDT / mass-drop
- **F3** primary Lund coordinates

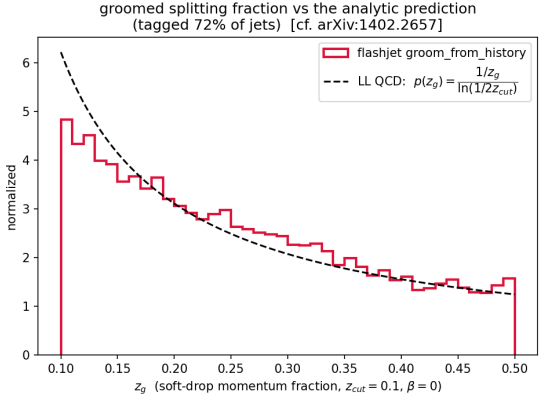
All three are pure-torch post-reads: no kernel changes, CPU/GPU identical, negligible cost.

Status

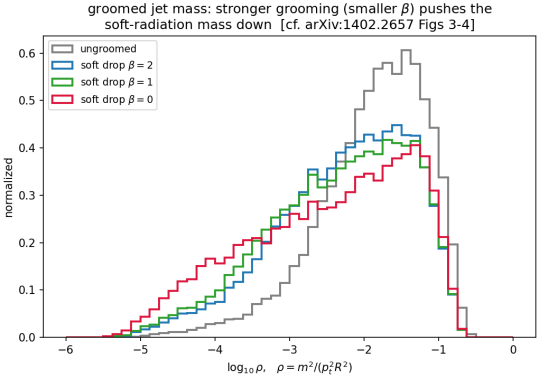
A **prototype** — the physics is validated and the speedups are real, but it is a Python/Triton library, **not** integrated into any experiment's reconstruction framework.

Correctness

The history is right: closures against analytic predictions



Soft-drop z_g follows the $1/z$ form predicted at leading-log, recovered from the C/A history.

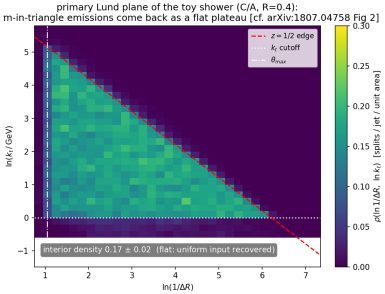


Groomed-mass ordering in β: larger β grooms less, so the mass peak moves up.

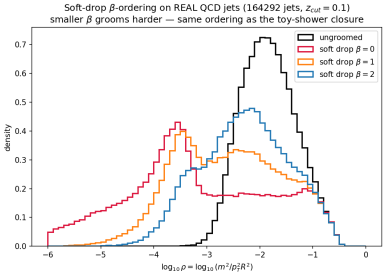
Toy-shower input, so the *prediction* is unambiguous — these test the decoders, not an event generator.

Clustering geometry

anti-k_t jet areas $\to \pi R^2$.



Lund plane uniform inside the kinematic triangle.



β-family ordering.

Independent NumPy tree-walks pin every decoder; the d-channel of the Lund output ties exactly to the splitting scales.

Same jets as FastJet — across the whole grid

3 algorithms $\times$ **5 radii** $\times$ **5 multiplicity bins** $\times$ **2 samples**, 20 000 jets per bin, on ATLAS Open Data.

check	**result**
number of jets found	**100.000 %** at every one of 150 points
leading-jet p_T within 10^{-4}	1.0000 at 148/150 points
median relative difference	$\sim 7 \times 10^{-8}$

k_t and C/A had never been checked against FastJet on real detector data — only against NumPy tree-walks. They now agree across the full radius $\times$ multiplicity grid.

The two exceptions are the *same single jet*: one constituent assigned across an R boundary. Jet counts still match and the leading jet is never ambiguous.

Same jets as the experiment

A stronger test than agreeing with FastJet.

ATLAS publishes an open dataset carrying both the **calorimeter clusters** and **its own reconstructed jets**. Cluster the ∼600 clusters per event ourselves, and compare:

n_{jets} vs ATLAS	**100.0 %** match
mean jets/event	10.96 vs 10.96

This closes the algorithm against **the experiment's own published output** — not against another implementation of the same algorithm.

599.3 clusters/event: the event regime, where the problem is $O(N^2)$.

Speed

The benchmark

Data — ATLAS Open Data, freely presentable

sample	jets	$\langle n_{\text{const}} \rangle$
boosted top	30 000	59.2
QCD (light q/g)	30 000	52.4

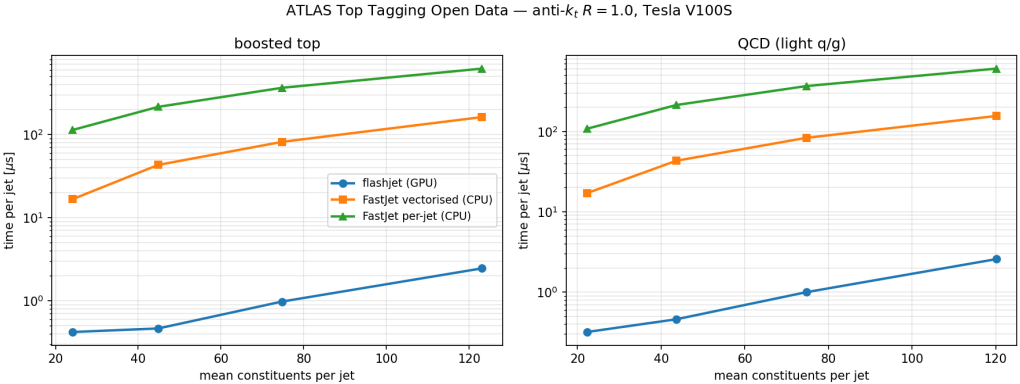
Large-R jets with calorimeter-cluster constituents.

Method

1. extract constituents once, save;
2. cluster on GPU, time it, save the *same* events;
3. cluster those saved events with FastJet;
4. **check they agree** — else the timing is meaningless.

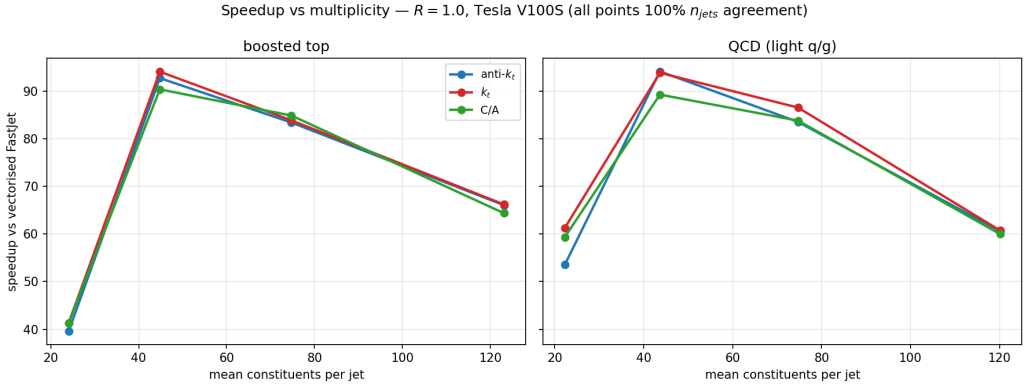
Both clusterers read the same saved file. Speedups are quoted against the **vectorised** FastJet interface — the faster, fairer baseline.

Time per jet



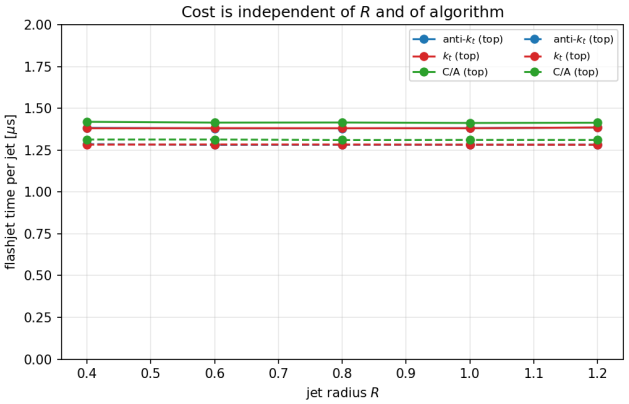
Log scale. flashjet stays nearly flat while both FastJet interfaces climb steeply with multiplicity.

The advantage grows with jet complexity



All three algorithms, both samples. Every point has **100 %** n_{jets} agreement. Busier jets → bigger win — the regime that matters for boosted-object physics.

Cost is independent of R — and of the algorithm



Across $R = 0.4 \to 1.2$: a **0.3 % spread** over a $3\times$ range in R.

R changes *which* pairs merge, not *how many* distances are computed.

Across algorithms: anti-k_t, k_t and C/A agree to within $\sim$4 % at every multiplicity.

k_t and C/A timing is **free**.

The numbers

sample	alg	$\langle n \rangle$	flashjet [μs/jet]	FastJet vec. [μs/jet]	speedup
top	anti-k_t	24.1	0.427	16.5	$39\times$
top	anti-k_t	44.8	0.464	43.0	$\mathbf{93}\times$
top	C/A	74.7	1.000	84.8	$85\times$
top	C/A	123.1	2.565	165.1	$64\times$
qcd	anti-k_t	22.2	0.346	17.5	$51\times$
qcd	anti-k_t	43.6	0.463	45.7	$\mathbf{99}\times$
qcd	anti-k_t	74.7	1.215	89.8	$74\times$
qcd	anti-k_t	120.2	2.673	154.9	$58\times$

39–99$\times$ over vectorised FastJet (median $64\times$), with 100 % jet-count agreement everywhere. Benchmarked on an **NVIDIA V100-class** GPU.

$R = 1.0$ rows shown. Against the per-jet FastJet interface the gap is roughly *two orders of magnitude*.

Caveats

- **The CPU baseline is FastJet's Python binding.** The production tool is C++. Until that is measured, the honest caveat is *"some of this could be Python overhead"* — the single most valuable missing test.
- **Large-R jets.** Boosted fat jets, $\sim$50–120 constituents — not small-R jets.
- **Batch regime.** The advantage comes from clustering *many* jets at once. A per-event framework may not exploit that — it needs measuring, not assuming.
- **Hardware.** A single GPU generation was benchmarked; these are not the best numbers the library can produce.

Conclusions

Summary

1. **Clusters on the GPU and keeps the full merge history**, so substructure — exclusive subjets, soft drop, Lund coordinates — comes free as array reads.
2. **Gives the same jets as FastJet.** 150/150 grid points at 100 % n_{jets} agreement, across anti-k_t/k_t/C-A, $R = 0.4$–1.2 and 24–123 constituents per jet. k_t and C/A validated on real data for the first time.
3. **Reproduces the experiment's own reconstructed jets** exactly, at ∼600 clusters per event.
4. **39–99× faster** than vectorised FastJet, and the advantage *grows* with jet complexity. Cost is independent of R and of algorithm.
5. Still a **prototype**, with meaningful headroom left.

What is next

	item	why
1	C++ FastJet baseline	every number here uses the Python binding; the production tool is C++
2	Tuning	the library has not been optimised for current hardware
3	Pile-up, grooming timing	untested axes a realistic workload will hit
4	Use inside experiment software	the real production test

Bottom line: as an accelerator-resident substitute for CPU clustering inside a learning workflow, flashjet is **correct** and roughly **two orders of magnitude faster** than per-jet CPU clustering.

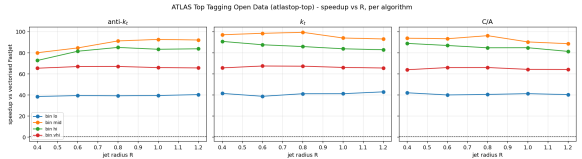
Thank you

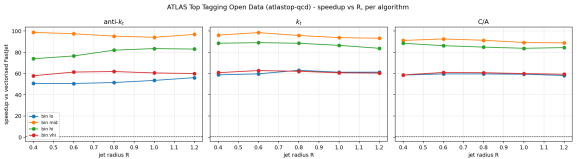
Questions?

Benchmarks: ATLAS Open Data (CC0).
Physics closures: analytic predictions on toy showers.

Backup

Backup — speedup vs R, per algorithm



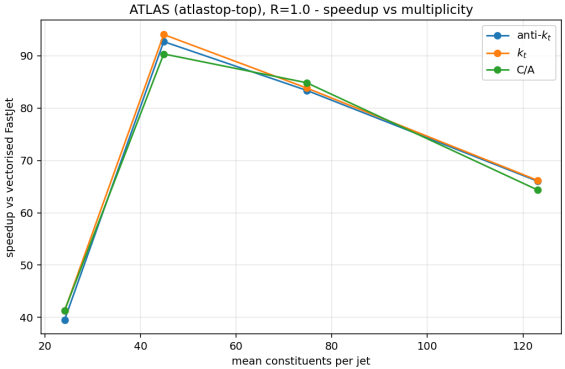


boosted top

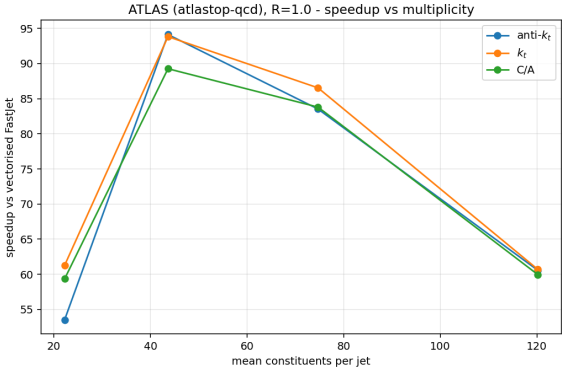
QCD (light q/g)

Flat in R for all three algorithms; ordering is by multiplicity bin.

Backup — speedup vs multiplicity



boosted top



QCD (light q/g)

Backup — the datasets

All benchmark data is **ATLAS Open Data** (CC0), read over the public redirector — no grid certificate required.

dataset	used for
ATLAS Top Tagging Open Data	timing + FastJet agreement
2020 ATLAS Jet Reconstruction	closure vs the experiment's own jets

Constituents are calorimeter clusters, massless. Physics closures use toy showers compared against leading-log analytic predictions, so the prediction being tested is unambiguous.